

BLUESTOCK FINTECH PVT. LTD.

Mutual Fund Analytics Platform

CAPSTONE PROJECT REPORT

End-to-End Data Engineering, ETL Pipeline & Interactive Dashboard

Company	Bluestock Fintech Pvt. Ltd.
Domain	Mutual Fund / Fintech
Project Type	Individual Capstone (1 Week)
Duration	7 Working Days ~50-55 Hours
Technologies	Python, SQL, Power BI, Pandas, Matplotlib
Submitted By	Prem Data Analyst Intern
Team Members	Sumit, Nikhita, Jyoti Prakash Gochhayat
Date	June 2026

40

Real Fund Schemes

10

Datasets Provided

87K+

Transaction Data

4.5 Yrs

NAV History

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1. Project Overview & Objective

Bluestock Fintech is a financial technology company focused on democratising investment analytics for retail and institutional investors in India. This capstone project builds a full-stack Mutual Fund Analytics Platform that ingests publicly available data from AMFI India, transforms it through a robust ETL pipeline, stores it in a relational database, and presents insights via an interactive dashboard.

1.1 Business Context

The Indian mutual fund industry has grown at a remarkable pace. As of December 2025, the industry manages over Rs. 81 lakh crore in AUM across 40+ schemes and 26.12 crore investor folios. Monthly SIP inflows crossed an all-time high of Rs. 31,002 crore in December 2025, reflecting India's deepening equity culture.

1.2 Project Objectives

#	Objective	Outcome
O1	Build ETL pipeline from raw AMFI data	Automated Python script
O2	Design normalised SQL schema for MF data	5-table star schema
O3	Perform comprehensive EDA on NAV & AUM	EDA notebook with 5+ charts
O4	Compute performance & risk metrics	Metrics dashboard
O5	Build interactive BI dashboard	HTML / Power BI file
O6	Analyse investor transaction patterns	Demographic insights
O7	Compare fund returns vs benchmark	Alpha / tracking error report
O8	Document and present the entire project	PDF report + slides

2. Problem Statement

Despite the massive growth of India's mutual fund industry, individual investors and financial advisors struggle to make data-driven fund selection decisions due to fragmented data, lack of unified analytics platforms, and complex financial metrics.

Problem	Description	Solution
P1: Data Fragmentation	NAV, AUM, SIP data on different platforms in different formats	ETL pipeline consolidating all data into single SQLite database
P2: Performance Gap	Cannot compare funds on risk-adjusted basis easily	Compute Sharpe, Alpha, Beta, VaR in unified dashboard
P3: No Benchmark Tracking	Retail investors unaware if fund beats its benchmark	Join NAV with benchmark index, compute rolling alpha
P4: Investor Blindspot	Limited visibility into demographic investment behaviour	Analyse transactions for demographic segmentation
P5: Slow Reporting	Monthly reports are static PDFs taking days to prepare	Interactive self-service dashboard refreshable from ETL output

3. Data Sources & Datasets

All datasets are derived from publicly available, freely accessible sources. No proprietary or confidential data is used.

3.1 Dataset Inventory

#	File	Rows	Description
01	fund_master.csv	40	Master list of 40 real AMFI schemes with codes, categories, expense ratios
02	nav_history.csv	~46,000	Daily NAV Jan 2022–May 2026, anchored to real mfapi.in values
03	aum_by_fund_house.csv	~90	Quarterly AUM for 10 fund houses from 2022–2025
04	monthly_sip_inflows.csv	48	Month-wise SIP inflow, accounts, registrations — real AMFI data
05	category_inflows.csv	~144	Net inflows by fund category for FY 2024-25
06	industry_folio_count.csv	21	Total MF folios broken by Equity, Debt, Hybrid
07	scheme_performance.csv	40	Sharpe, Sortino, Alpha, Beta, Max Drawdown per fund
08	investor_transactions.csv	~32,000	SIP/Lumpsum/Redemption for 5,000 investors with demographics
09	portfolio_holdings.csv	~320	Top equity holdings with sector weights per fund
10	benchmark_indices.csv	~8,000	Daily Nifty 50, Nifty 100, BSE SmallCap, CRISIL indices

3.2 Key Real-World Data Points

Metric	Value	Source
SBI MF AUM Dec 2025	Rs. 12.50 lakh crore (largest AMC)	AMFI Quarterly
ICICI Pru MF AUM Dec 2025	Rs. 10.74 lakh crore	AMFI Quarterly
HDFC MF AUM Dec 2025	Rs. 9.30 lakh crore	AMFI Quarterly
SIP Inflow Dec 2025	Rs. 31,002 crore — all-time high	AMFI Monthly Note
Active SIP Accounts Dec 2025	9.35 crore accounts	AMFI Monthly Note
Total MF Folios Dec 2025	26.12 crore	AMFI / Business Standard
Industry AUM Dec 2025	Rs. 81 lakh crore	AMFI

4. System Architecture & ETL Pipeline

The project follows a classic data engineering architecture: Extract → Transform → Load → Analyse → Visualise. This mirrors real-world fintech data pipelines used at companies like Zerodha, Groww, and Paytm Money.

Layer	Component	Tools
Layer 1: Extract	10 CSV files from AMFI India; mfapi.in REST API	Python requests, pandas
Layer 2: Transform	Clean, validate, forward-fill NAV; compute daily returns	Pandas, NumPy
Layer 3: Load	5-table star schema in SQLite	SQLAlchemy, sqlite3
Layer 4: Analyse	EDA, risk metrics, benchmark comparison, cohort analysis	Matplotlib, Seaborn, SciPy
Layer 5: Visualise	Interactive 6-page HTML dashboard	Chart.js, HTML/CSS

4.1 Database Schema (Star Schema)

Table	Type	Key Columns	Rows
dim_fund	Dimension	amfi_code (PK), fund_house, category, expense_ratio	40
dim_date	Dimension	date_id (PK), date, year, month, quarter	1,500
fact_nav	Fact	amfi_code (FK), date (FK), nav, daily_return_pct	46,000
fact_transactions	Fact	tx_id (PK), investor_id, amfi_code (FK), amount, type	32,000+
fact_performance	Fact	amfi_code (FK), sharpe, alpha, max_dd, score	40
fact_portfolio	Fact	amfi_code (FK), stock_symbol, weight_pct, sector	320
fact_aum	Fact	fund_house, date, aum_crore, num_schemes	90
fact_sip_industry	Fact	month, sip_inflow_crore, sip_accounts_crore	48

5. 7-Day Task Breakdown

Day	Focus	Key Tasks	Deliverables
Day 1	Project Setup + Data Ingestion	Load 10 CSVs; fetch live NAV from mfapi.in; validate AMFI codes	data_ingestion.py, raw CSVs
Day 2	Data Cleaning + SQL Design	Clean NAV/transactions; design star schema; load SQLite	bluestock_mf.db, schema.sql, queries.sql
Day 3	Exploratory Data Analysis	5+ charts: NAV trend, AUM, SIP, age groups, state volume	EDA notebook, chart PNGs
Day 4	Performance Analytics	CAGR, Sharpe, Sortino, Alpha, Beta, Max DD, Scorecard	fund_scorecard.csv, performance notebook
Day 5	Dashboard Development	6-page interactive HTML dashboard with slicers	bluestock_mf_dashboard.html
Day 6	Advanced Analytics	VaR, CVaR, cohort analysis, SIP continuity, recommender	advanced_analytics.ipynb
Day 7	Final Report + GitHub	Write report, create slides, clean code, push to GitHub	Final_Report.pdf, Presentation.pptx

6. Exploratory Data Analysis (EDA) Findings

Comprehensive EDA was performed across all 10 datasets. Five publication-quality charts were generated and saved to the reports/ folder.

6.1 Charts Generated

- Chart 1 — NAV Trend (2022–2026): Daily NAV plotted for top 5 funds. All equity funds show strong recovery post-2022 correction with SBI Small Cap outperforming consistently.
- Chart 2 — AUM by Fund House: SBI Mutual Fund dominates with Rs. 12.5 lakh crore, followed by ICICI Prudential (Rs. 10.74L Cr) and HDFC (Rs. 9.3L Cr).
- Chart 3 — SIP Inflow Trend: Monthly SIP inflows grew 169% from Rs. 11,517 crore (Jan 2022) to Rs. 31,002 crore (Dec 2025), a steady uninterrupted growth.
- Chart 4 — Investor Age Distribution: The 36-45 age group is the most active. Investors aged 56+ have the highest average SIP amount of Rs. 11,575.
- Chart 5 — Transaction Volume by State: Punjab, Tamil Nadu and Madhya Pradesh lead in transaction volume. T30 cities contribute 65.8% vs B30 at 34.2%.

6.2 Key EDA Insights

- Folio count doubled from 13.26 crore (Jan 2022) to 26.12 crore (Dec 2025) — 97% growth in 4 years.
- Banking and IT sectors together constitute 34% of all equity portfolio weight.
- 92% of investors are KYC verified; 8% remain pending.
- Lumpsum transactions account for 60% of total volume vs SIP at only 6%.
- Liquid fund category recorded the highest net inflows at Rs. 4,51,275 crore.

7. Fund Performance Analytics

All key performance and risk metrics were computed from raw NAV history using Python (Pandas, NumPy, SciPy). A composite scorecard ranks all 40 schemes.

7.1 Metrics Computed

Metric	Formula	Interpretation
CAGR	$(NAV_end / NAV_start)^{(1/years)} - 1$	Annualised compound growth rate
Sharpe Ratio	$(R_p - R_f) / \sigma_p \times \sqrt{252}$, $R_f = 6.5\%$	Risk-adjusted return vs risk-free rate
Sortino Ratio	$(R_p - R_f) / \sigma_{down} \times \sqrt{252}$	Penalises only downside volatility
Alpha	OLS intercept $\times 252$ vs Nifty 100	Return above benchmark
Beta	OLS slope vs Nifty 100	Market sensitivity
Max Drawdown	$\min(NAV / running_max - 1)$	Worst peak-to-trough decline
VaR (95%)	5th percentile of daily returns	Daily loss at 95% confidence
CVaR (95%)	Mean of returns $\leq$ VaR threshold	Expected loss beyond VaR
Composite Score	30% CAGR + 25% Sharpe + 20% Alpha + 15% Expense + 10% DD	Fund ranking score

7.2 Top 8 Funds by Composite Score

#	Scheme	Category	3yr %	Sharpe	Alpha	Max DD	Score
1	SBI Small Cap - Direct	Small Cap	23.1%	0.93	0.98	-13.1%	72.8
2	Kotak Flexicap - Regular	Flexi Cap	15.7%	0.98	1.56	-16.2%	71.5
3	Kotak Emerging Equity	Mid Cap	18.2%	0.96	1.67	-17.8%	70.5
4	ABSL Small Cap - Reg	Small Cap	22.4%	0.90	0.22	-14.1%	68.3
5	Mirae Asset Large Cap	Large Cap	14.8%	1.06	3.27	-19.4%	63.9
6	ICICI Pru Liquid - Reg	Liquid	7.7%	7.68	0.12	-0.1%	61.8
7	ICICI Pru Bluechip - Dir	Large Cap	14.4%	1.03	2.89	-20.1%	59.6
8	HDFC Short Term Debt	Debt	7.4%	1.84	0.87	-3.2%	63.0

7.3 Key Performance Findings

- SBI Small Cap Direct scores highest (72.8/100) with 23.1% 3yr CAGR and Sharpe ratio of 0.93.

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- Liquid funds dominate Sharpe ratio — ICICI Pru Liquid scores 7.68 due to near-zero volatility.
 - Mirae Asset Large Cap has the best Alpha (3.27%) indicating strongest outperformance vs Nifty 100.
 - All equity funds have Beta close to 1.0, confirming strong correlation with market benchmark.
 - ABSL Small Cap has the worst Max Drawdown at -14.1%, while Liquid funds are near zero.

8. Advanced Analytics

8.1 Value at Risk (VaR) & CVaR

Historical VaR (95%) and CVaR were computed for all 40 funds from daily NAV returns.

Fund	VaR 95%	CVaR 95%	Risk Level
ABSL Small Cap	-2.60%	-3.25%	Very High
UTI Mid Cap	-1.92%	-2.33%	High
SBI Small Cap Direct	-1.87%	-2.34%	High
HDFC Top 100	-1.44%	-1.81%	Moderate
HDFC Short Term Debt	-0.38%	-0.50%	Low
ABSL Liquid Fund	-0.03%	-0.04%	Very Low

8.2 Investor Cohort Analysis

Investors were grouped by their first transaction year to study retention and investment behaviour.

- 2022 cohort: Earliest investors with highest average portfolio value.
- 2024-2025 cohort: Newer investors tend to start with smaller SIP amounts but grow quickly.
- At-risk SIP investors (gap > 35 days): Flagged for outreach campaigns.

8.3 Fund Recommender

A simple rule-based recommender was built that filters funds by SEBI risk grade and ranks by Sharpe ratio.

Risk Appetite	Eligible Grades	Top Recommendation
Low	Low only	ICICI Pru Liquid Fund (Sharpe: 7.68)
Moderate	Low, Moderate	Mirae Asset Large Cap (Sharpe: 1.06)
High	Low, Moderate, High	Kotak Emerging Equity (Sharpe: 0.96)
Very High	All grades	SBI Small Cap Direct (Score: 72.8)

9. Deliverables & Evaluation

ID	Deliverable	Format	Weight	Criteria
D1	ETL Pipeline Script	.py	15%	Runs without manual steps, error handling, clean code
D2	SQLite Database	.db	10%	Correct schema, all data loaded, queries run correctly
D3	EDA Notebook	.ipynb	15%	Depth of analysis, chart quality, insights documented
D4	Performance Metrics	.ipynb + CSVs	15%	Mathematical accuracy, correct Sharpe/Beta/VaR formulas
D5	Interactive Dashboard	.pbix / .html	20%	Design quality, all 4 pages, slicers functional
D6	Advanced Analytics	.ipynb	10%	VaR correctness, cohort analysis, recommender logic
D7	Final Report + Slides	.pdf + .pptx	15%	Professional quality, complete sections, clear findings

9.1 Technical Stack

Category	Tool	Purpose
Language	Python 3.10+	All data processing and ETL
Data Manipulation	Pandas 2.0+	DataFrames, cleaning, aggregation
Numerical	NumPy	Risk metrics, statistical functions
Visualisation	Matplotlib, Seaborn	Static charts for reports
Dashboard	Chart.js (HTML)	Interactive 6-page dashboard
Database	SQLite3 + SQLAlchemy	Local relational star schema
Statistics	SciPy	OLS regression for Beta/Alpha
Notebooks	Jupyter Lab	EDA and analytics (5 notebooks)
Version Control	Git + GitHub	Code versioning and submission

10. Key Findings & Insights

The following are the most significant findings from this analytics project:

#	Finding
1	SBI Mutual Fund dominates the industry with Rs. 12.5 lakh crore AUM — it has more than doubled since 2022.
2	SIP inflows grew 169% over 4 years, hitting Rs. 31,002 crore all-time high in December 2025.
3	SBI Small Cap Direct tops the composite scorecard (72.8/100) with 23.1% 3yr CAGR.
4	Banking (20%) + IT (14%) = 34% of all equity portfolio weight across schemes.
5	ABSL Small Cap has the highest daily VaR at -2.60% vs. Liquid funds at -0.03%.
6	Investors aged 56+ have the highest average SIP amount at Rs. 11,575 per transaction.
7	T30 cities contribute 65.8% of transaction volume; B30 cities growing.
8	Folio count doubled from 13.26 crore (2022) to 26.12 crore (2025) — 97% growth.
9	Mirae Asset Large Cap has the best Alpha (3.27%), consistently beating Nifty 100.
10	92% of all 5,000 investors are KYC verified; compliance is high across demographics.

11. Limitations & Future Work

11.1 Current Limitations

- Investor transaction data is synthetic — real AMFI investor-level data is not publicly available.
- Dashboard does not auto-refresh from live API; requires manual ETL re-run.
- Portfolio holdings are only for December 2025 — no historical holdings data available.
- Only 40 schemes covered out of 1,908+ available on AMFI.
- Power BI .pbix file requires manual connection to CSV files.

11.2 Future Enhancements

- B1: Schedule ETL as automated cron job fetching NAV from mfapi.in daily at 8 PM.
- B2: Build Streamlit web app as alternative to Power BI for real-time dashboard.
- B3: Monte Carlo simulation projecting NAV growth over 5 years with uncertainty bands.
- B4: Markowitz Efficient Frontier portfolio optimisation for 5 selected funds.
- B5: Automated HTML email report generator sending weekly performance summaries.
- Expand to all 1,908 AMFI schemes with PostgreSQL for production-grade database.
- Add real-time WebSocket integration for live NAV updates in dashboard.

12. Conclusion

This capstone project successfully demonstrates the complete lifecycle of a real-world data engineering project — from raw data ingestion to an interactive analytics dashboard. The platform covers 40 AMFI-registered mutual fund schemes, processes over 46,000 NAV records and 32,000 investor transactions, and delivers actionable insights through 6 dashboard pages and 5 Jupyter notebooks.

The project addresses real business problems faced by the Indian mutual fund industry: data fragmentation, lack of risk-adjusted performance comparison, and slow reporting. The ETL pipeline, SQLite database, performance metrics, and interactive dashboard together form a complete, reproducible analytics system that can be extended to production scale.

Key technical skills demonstrated include: Python data engineering, SQL database design, financial mathematics (Sharpe, Alpha, Beta, VaR), exploratory data analysis, interactive dashboard development, and professional project documentation.
